



MILL TEST CERTIFICATE

XIS PIPE AND TUBE, LLC
451 Louis E. Mikulin Rd.
ryan TX 77807

Customer:
MRC Global
PO Box
Charleston, WV, 25322
877-294-7574

Sales Order:
0630004951
PO:
S 7AA 637379

Invoice No.:
1800049629

Date of Issue:
07/23/2024

Certificate No.:	1600049629-4
Product Description:	SL 12 750 0 500 65 48# X52BDRHGBW-20P2
Specification:	SL 46th Edition May 2019

Test Number	Lot Number	Mat Number	
B403191	L724185030	332758	Steel Making / Coil Rolling Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US

Tensile Test - Strip Specimen Gauge Length 2"										Mechanical Properties												
Temp (F) -20							CVN			Temp (F) -20												
Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRA)	Body (ft-lbf)			Shear Area (%)			Weld (ft-lbf)			Shear Area (%)			Size	Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)
T180°	1 1/2"	63500	72300	75400	0.88	43.0	82	1	2	3	1	2	3	1	2	3	1	2	3	10x7.5 mm	3T90	15
								Avg			Avg			Avg			Avg					
								UB	UB	UB	UB	UB	UB	UB	UB	UB	UB	UB	UB			
								Lateral Expansion (in)						Lateral Expansion (in)								

Analysis		Chemical Composition (Max %)																	
[H/P]	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (401 Max. Ac.)	CE Results	
H	0.06	0.210	0.88	0.010	0.002	0.05	0.06	0.010	0.12	0.002	0.017	0.028	0.015	0.0004	0.0019	0.0069	0.25	0.12	
P	0.05	0.192	0.83	0.008	0.000	0.05	0.05	0.010	0.12	0.003	0.015	0.023	0.014	0.0003	0.0024	0.0054	0.25	0.11	
P	0.05	0.190	0.83	0.008	0.000	0.05	0.05	0.010	0.12	0.002	0.017	0.023	0.014	0.0003	0.0023	0.0053	0.25	0.11	

Following Controls / Tests have been satisfactorily performed:

1. - Hydrostatic Testing:	4,090.00 psi Pressure for 5.00 seconds
2. - Ultrasonic Tested to N10	- passed
3. - Seam Annealed min 1,600 F	
4. - Flattening Test Passed	

acronyms:
 L = Longitudinal
 T = Transverse
 Y.S. = Yield Strength
 E.L. = Elongation
 P.A. = Product Analysis
 U.B. = Un-Broken
 T = Transverse
 U.T.S. = Ultimate Tensile Strength
 H = Heat Analysis
 A.C. = Acceptance Criteria
 N/A = Not Applicable

Remarks:

1. This material meets: Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2. This product meets the requirements of ASME BPVC, II A Specification SA-53B-2023
3. Made and Melted in USA
4. Date of Manufacture 07/2024
5. This product is silicon and/or aluminum fully killed.
6. Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7. Meets Hardness requirements of NACE MR0175-2021 / ISO 15156:2020 Clause A.2.1.2.

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

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Quality Manager